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PI Name:	Limoli, Charles Ph.D.		
Project Title:	NSCOR: Mechanisms Underlying Charged Particle-Induced Disruption of CNS Function		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
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Key Personnel Changes/Previous PI:			
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Task Description:	Humans exposed to the space radiation environment encounter a different type of radiation than that found on Earth. Highly energetic particles can traverse the body, including the brain, and leave a path of damage that is difficult to repair and disrupts function. One consequence of this exposure is the development of radiation-induced cognitive dysfunction. The focus of this application is to resolve the mechanisms leading to radiation-induced cognitive dysfunction, in order to improve risk estimates for early and late CNS (central nervous system) effects caused by exposure to the space radiation environment. We have found that radiation impacts excitatory and inhibitory circuitry in the brain differently, and leads to significant disruptions to the structural integrity of neurons while altering the levels and activity of the proteins that regulate neuronal function. The situation is further complicated by the fact that different regions of the brain exhibit differential sensitivity to charged particle irradiation. Based on the foregoing we propose a comprehensive series of mechanistic studies analyzing the biochemical, structural, and electrophysiologic changes induced by radiation exposure that alter function to impact cognition. Carefully selected cognitive tasks and a wide variety of approaches will be used to critically analyze microcircuits in the brain and how charged particle-induced changes are transmitted via unique signaling pathways between cells that can alter gene expression profiles within cells. Collectively these studies will span a radiation effects pathway from synapse to cognition, with an overarching goal of determining whether charged particle irradiation leads to alterations in the excitatory and inhibitory activity of the brain at a global and/or region specific level. Our studies will therefore elucidate the functional consequences of radiation-induced changes on cognition that impose risk on mission critical activities and long-term cognitive health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Work will characterize several of the adverse effects of ionizing radiation on the CNS and help to define the early and late CNS risks associated with exposure to the space radiation environment. There is a real possibility that many of the consequences of cosmic radiation exposure will extend over many years and impact (possibly negatively) neurocognitive health after return to Earth. Thus, research elucidating the mechanistic basis of charged particle-induced effects in the irradiated brain may well have functional importance to terrestrial situations where the public may be exposed to ionizing radiation from a variety of potential sources.
	The overarching hypothesis remains unchanged and still seeks to define the mechanisms contributing to charged particle–induced disruptions to CNS functionality that impact risk over acute (mission critical) and more chronic (terrestrial) timeframes. Despite the changes implemented to the team, our group has made some of the most significant progress to data, finalizing many data sets utilizing mixed beams simulating the GCR (galactic cosmic radiation), low dose rate exposures using neutrons, along with new behavioral paradigms designed to track human behavioral performance more precisely. Our studies continue to evaluate the neurological consequences of cosmic radiation exposure from the synapse to multiple neurocognitive domains. Work using the lighter, more abundant protons and He ions continues to reveal the remarkable persistence of adverse effects over the course of a 1-year post-irradiation interval. In these studies, a wealth of neurobehavioral decrements has been found to coincide with long-range disruptions in circuit connectivity across multiple brain regions, changes that also transpire amidst significant increases in neuroinflammation. We have now completed to “acute – 1 day” GCR simulations (+/- the presence of 56Fe). In the absence of Fe, we have significant behavioral decrements along with paired cell recordings that indicate changes in certain intrinsic excitability between pyramidal cells in the CA1 region of the hippocampus. We are also finalizing our data sets detailing significant differences between the sexes in regards to their neurobehavioral responses to cosmic radiation exposure. A theme championed by Dr. Britten, namely, to increase the rigor of behavioral testing in efforts to uncover more subtle deficits, possibly more relevant to human behavioral outcomes has now been implemented in earnest. Animal cohorts subjected to the mixed beam GCR have now been evaluated on a novel location memory (NLM)/updating task that relies on an animal to progressively update their prior memory of an object moved in sequence over time. This task has been shown to have strong cross-species parallels with human behavioral testing and has now shown that the mixed field GCR leads to significant decrements on this task, 6 months post-irradiation. This exciting data set has also revealed reductions in long-term potentiation (LTP) measured from the hippocampus and medial prefrontal cortex, changes that can be functionally reversed using a histone deacetylase 3 (HDAC3) inhibitor. These data point to a continued theme in that increasing the rigor of behavioral testing invariably uncovers radiation-induced deficits, which may confound in flight mission critical performance. Data also begins to point to additional countermeasures that NASA may consider for combating the adverse effects of space radiation exposure. Work at Stanford has now been coordinated between Eastern Virginia Medical School (EVMS) and University of California, Irvine (UCI) and has linked electrophysiological changes to neurobehavioral and molecular alterations after exposure to various charged particles. Paired cell recordings, live animal imaging, and advanced field recording have now been undertaken and have uncovered some fascinating changes in both intrinsic cell properties and network connectivity that correlate with changes in behavior. Work from EVMS continues to uncover an impressive array of executive function deficits at the inter-individual level in pre-trained rats and importantly, through implementation of additional testing designed to increase the stringency of cognitive tasks (UC-Flex, risk aversion). This theme continues to point to the elevated CNS risks associated with cosmic radiation exposure, as irradiated rats asked to execute increasingly demanding tasks (i.e., abstract reasoning) continue to exhibit significant decrements compared to their unirradiated cohort. Many of these changes have now been linked to layer specific deficits implicating changes in both LTP and long-term depression (LTD) in the medial prefrontal cortex (mPFC), changes that may also portend those changes seen in mice exhibiting reduced serotonin levels in the dorsal hippocampus. Dr. Britten has also been the driver for implementing a novel statistical approach involving kernel density estimation (KDE), able to calculate the proportion of mice that exhibited impaired performance within each cohort for a given behavior task or other CNS measure (i.e., electrophysiology). This has helped the NASA Specialized Center of Research (NSCOR) team tremendously by providing a tangible risk estimate that helps bridge the gap between experimental statistics and bona-fide risk estimation. Lastly, and perhaps most relevant to NASA’s overall mission to define radiation risks relevant to actual mission deep travel, we have completed extensive analyses of low dose rate neutron/gamma exposed animals. Mice exposed to 1 mGy/day over a six-month interval have been shown to exhibit significant impairments on a range of independent behavioral tasks spanning multiple open field testing, social interaction, anxiety, and fear extinction. Importantly, behavioral changes were directly linked to reduced LTP in the hippocampus and mPFC and altered cell intrinsic properties. These data reveal for the first time, that low dose rate exposures designed to replicate the space radiation

Task Progress:	environment, elicit long-lasting and severe impairments in CNS functionality, again pointing to the elevated risks associated with deep space travel. Many of these highlights are detailed below. Research Highlights for the CNS NSCOR: Work under the NSCOR has been able to demonstrate direct links between charged particle induced disruptions to cognition and persistent damage to specific neuronal subsets interrogated by our behavioral tasks. In addition to published findings, we have now generated some uncovered fascinating effects of GCR and low dose rate exposures that point further to the elevated risks associated with deep space radiation exposure. Other work extends many of our structural plasticity studies to ultra-high resolution electron microscopy, and demonstrates now for the first time how low dose charged particles reduced synaptic density and axonal myelination within the brain. These studies are now published (Ref. 1 below and included in Bibliography section). Follow up studies with Dr. Cucinotta have developed further stochastic modeling approaches for analyzing and predicting radiation-induced damage to dendritic structures in the brain. These studies can be extended to charged particle exposures and are also published (Ref. 2 below). 1. Dickstein, D.L., Talty, R., Bresnahan, E., Varghese, M., Perry, B., Janssen, W.GM., Sowa, A., Giedzinski, E., Apodaca, L, Baulch, J.E., Acharya, M.A., Parihar, V. and Limoli C.L. Alterations in synaptic plasticity and myelination in response to exposure to high-energy charged particles. J. Comparative Neurology doi.org/10.1002/cne.24530 (2018). 2. Cacao, E., Parihar, V., Limoli. C.L., and Cucinotta F.A. Stochastic modeling of radiation-induced dendritic damage on in silico mouse hippocampal neurons. Sci. Rep. 8(1):5494. doi: 10.1038/s41598-018-23855-9 (2018). A collaborative effort between all NSCOR Principal Investigators (PIs) involving a comprehensive assessment of low dose rate neutron exposures has led to the following manuscript currently under review: 3. Acharya, M.M., Baulch, J.E., Klein, P.M., Baddour, A.A.D., Apodaca, L.A., Kramár, E.A., Alikhani, L., Garcia Jr. C., Angulo, M.C., Batra, R.S., Fallgren, C.M., Borak, T.B., Stark, C.E.L., Wood, M.A., Britten, R.A., Soltesz, I. and Limoli, C.L. New concerns for neurocognitive function during deep space exposures to chronic, low dose rate, radiation. Submitted, eNeuro, (2019). A manuscript detailing the benefits of microglial elimination for the preservation of neurocognitive function after cosmic radiation exposure is also in preparation: 4. Allen, B.D., Syage, A.R., Moroso, M., Baddour, A.A.D., Luong, V., Minasyan, H., Giedzinski, E., West, B.L., Baulch, J.E., Soltesz, I., Limoli, C.L. and Acharya, M.M. Mitigation of 4He irradiation-induced brain injury by microglia depletion. Manuscript in Preparation. Work at Stanford has produced achieved some Major milestones and accomplishments: • Determined that chronic neutron irradiation reduces the ability of mouse hippocampal neurons to appropriately respond to stimuli, due to decreases in their resting membrane potential and an increased rheobase current being required to evoke action potential generation. • Observed that chronic neutron irradiation suppresses the frequency of excitatory synaptic signaling among hippocampal neurons in mice, which undermines important learning and memory functions. • Established that space-relevant doses and compositions of galactic cosmic radiation induce excessive inhibitory synaptic signaling among hippocampal neurons in mice, thus disrupting normal circuit function. • Discovered that space-relevant doses and compositions of galactic cosmic radiation disrupt the precise rhythmic interactions of neurons in the hippocampus of behaving mice, likely underlying associated deficits in learning and memory behavior. Work with rats between EVMS and Loma Linda University (LLU) has revealed some interesting effects in long-term depression that is specific to layers (Layer 5) within the pre-frontal cortex. This work has now progressed and a manuscript is nearly ready for submission. 5. Richard A. Britten, Vania D. Duncan, Arriyam Fesshaye, Emil Rudobeck, Evangeline Macadat, Jessica S. Jewell, Abigail Tondin, Gregory A Nelson and Roman Vlkolinsky. Altered cognitive flexibility and synaptic plasticity in the rat prefrontal cortex following exposure to low (=15 cGy) doses of 28Si particles. Manuscript in preparation. In summary, much of our ongoing studies have added to our knowledge of space radiation effects on the brain, and have now laid additional groundwork for some exciting upcoming studies utilizing full spectrum “33-mixed beam” GCR simulations, additional low dose rate studies with neutrons at Colorado State University, Pion exposures, in vivo recordings of specific neuronal ensembles, and neuroproteomics linked to epigenetic changes in the brain. Lastly, we will embark on a major countermeasure study at the upcoming NASA Space Radiation Laboratory (NSRL) NSRL19B run at Brookhaven National Laboratory (BNL).
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Dickstein DL, Talty R, Bresnahan E, Varghese M, Perry B, Janssen WGM, Sowa A, Giedzinski E, Apodaca L, Baulch J, Acharya M, Parihar V, Limoli CL. "Alterations in synaptic plasticity and myelination in response to exposure to high-energy charged particles." J Comp Neurol. 2018 Dec 1;526(17):2845-55. https://doi.org/10.1002/cne.24530 ; PubMed PMID: 30198564; PubMed Central PMCID: PMC6442721 , Dec-2018
Articles in Peer-reviewed Journals	Baulch JE. "Radiation-induced genomic instability, epigenetic mechanisms and the mitochondria: A dysfunctional ménage a trois?" Int J Radiat Biol. 2019 Apr;95(4):516-25. Epub 2018 Dec 10. https://doi.org/10.1080/09553002.2018.1549757 ; PubMed PMID: 30451575 , Apr-2019